Day 1

1) What is the Web?

The **web** is a system of interconnected pages and applications that you access through the internet using a browser. It includes websites, web pages, and web apps that are linked through URLs and hosted on servers. The web is one part of the internet, but not the whole internet.

Website

A **website** is a collection of related web pages grouped under one domain name. For example, a news site, a school site, or a company site can have many pages like Home, About, Contact, and Services. Websites are often mainly used to display information.

Web page

A **web page** is a single page on the web. It can contain text, images, links, videos, forms, and other content. For example, the "About Us" page on a company website is one web page.

Web app

A **web application** is a website that behaves like software and lets users interact with it. Examples include Gmail, Google Docs, YouTube, online banking, and shopping carts. Web apps usually let users log in, enter data, and perform tasks rather than just read information.

2) What is the Internet?

The **internet** is a global network of connected computers and devices that communicate with each other. It is the infrastructure that makes online communication, websites, email, streaming, and file transfer possible. The web runs on top of the internet, but the internet also supports many other services besides the web.

IP address

An **IP address** is a unique number used to identify a device or server on the internet. It works like a digital address so data can be sent to the correct place. Devices and servers need IP addresses to find and communicate with one another.

DNS

DNS stands for Domain Name System. It translates human-friendly domain names like `google.com` into machine-friendly IP addresses like `142.250...` so browsers know where to go. DNS is often called the internet's phonebook or address book because it helps match names to numbers.

3) Search engine and browser

A **browser** is an application used to open and view web pages and web apps. Common examples are Google Chrome, Mozilla Firefox, Microsoft Edge, Safari, and Opera. A browser sends requests to websites, receives web content, and displays it in a readable form.

A **search engine** is a service that helps you find information on the web by searching indexed web pages. Common examples are Google, Bing, DuckDuckGo, Yahoo Search, and Brave Search. You use a search engine inside a browser to search the web more easily.

Difference between browser and search engine

- A **browser** is the tool you use to access websites.
- A **search engine** is the tool you use to find websites.
- Example: Chrome is a browser, while Google Search is a search engine.

4) Programming language vs markup language

A **programming language** is used to write instructions that a computer can execute to perform tasks or solve problems. Examples include Python, Java, C, C++, and JavaScript. These languages can control logic, conditions, loops, calculations, and behavior.

A **markup language** is used to structure and describe content, not to perform logic like a programming language. It uses tags to organize text and data so that software can understand how content should be presented. Examples include HTML and XML.

Main differences

- Programming language gives instructions to the computer.
- Markup language describes and organizes content.
- Programming language can perform logic and operations.
- Markup language mainly focuses on structure and presentation.

5) Difference between HTML, XML, and CSS

These three are related to web development, but they have different jobs.

HTML

HTML stands for HyperText Markup Language. It is used to create the structure of web pages, such as headings, paragraphs, links, images, buttons, and forms. HTML tells the browser what content exists on the page.

XML

XML stands for Extensible Markup Language. It is used to store and transport data in a structured way. XML focuses on describing data clearly rather than displaying it on a screen. It is often used in configuration files, data exchange, and APIs.

CSS

CSS stands for Cascading Style Sheets. It is used to style and design web pages. CSS controls colors, fonts, spacing, layout, and responsiveness. If HTML is the structure, CSS is the appearance.

Simple comparison

Technology	Purpose	Main use
HTML	Structure	Builds the content of a web page
XML	Data storage/exchange	Organizes data in a readable format
CSS	Styling	Makes the page look good

Easy example

- HTML: creates a heading and paragraph.
- XML: stores information like name, age, and city.
- CSS: changes the heading color, font size, and spacing.